



Bibliography

1. Aavatsmark, I., Barkve, T. and Mannseth, T. "Control-Volume Discretization Methods for 3D Quadilateral Grids in Inhomogeneous, Anisotropic Reservoirs," paper SPE 38000, *Society of Petroleum Engineers Journal* (1998) 3, No. 2, 146-154; also presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (June 1997).
2. Abbott, M.M. and van Ness, H.G. *Thermodynamics with chemical applications*, Schaum's Outline Series, USA, McGraw-Hill (1989).
3. Ansari, A. M., Sylvester N. D., Sarica O., and Shoham J. P. "A Comprehensive Mechanistic Model for Upward Two-Phase Flow in Wellbores," *SPE Production & Facilities* (1994) 9, No. 2, 143-151.
4. Appleyard, J.R. and Cheshire, I. M. "Nested Factorization," paper SPE 12264, presented at the SPE Symposium on Reservoir Simulation, San Francisco, California, USA (November 15-18, 1983).
5. Arri, L.E., Yee, D., Morgan, W.D., and Jeansonne, N.W. "Modeling Coalbed Methane Production with Binary Gas Sorption," paper SPE 24363, presented at the SPE Rocky Mountain Regional Meeting, Casper, Wyoming, USA (May 18-21, 1992).
6. Aziz, K., Govier, G. W., and Fogarasi, M. "Pressure Drop in Wells Producing Oil and Gas," *Journal of Canadian Petroleum Technology* (1972) 11, No. 3, 38-48.
7. Aziz, K., Ramesh, A. B., and Woo, P. T. "Fourth SPE Comparative Solution Project: Comparison of Steam Injection Simulators," *Journal of Petroleum Technology* (1987) 39, No. 12, 1576-1584.
8. Aziz, K. and Settari, A. *Petroleum Reservoir Simulation*, London, UK, Applied Science Publishers (1979) 398.
9. Baker, L. E. "Three-Phase Relative Permeability Correlations," paper SPE 17369, presented at the SPE Enhanced Oil Recovery Symposium, Tulsa, Oklahoma, USA (April 16-21, 1988).
10. Bardon, C. and Longeron, D. G. "Influence of Very Low Interfacial Tensions on Relative Permeabilities," paper SPE 7609, *Society of Petroleum Engineers Journal* (1980) 20, No. 15, 391-401.
11. Barker, J.W. and Fayers F.J. "Transport Coefficients for Compositional Simulation With Coarse Grids in Heterogeneous Media," paper SPE 22591, *SPE Advanced Technology Series* (1994) 2, No. 2, 103-112.
12. Beattie, C.I., Boberg, T.C., and McNab, G.S. "Reservoir Simulation of Cyclic Steam Stimulation in the Cold Lake Oil Sands," paper SPE 18752, *Society of Petroleum Engineers Journal*, (1991) 6, No. 2, 200-206.
13. Beggs, H. D. *Production Optimization Using Nodal Analysis*, OGCI Publications, Oil & Gas Consultants Inc., Tulsa, USA (1991) 81.

14. Bowers, B.E., Brownlee, R.F., and Schrenkel, P.J. "Development of a Downhole Oil/Water Separation and Reinjection System for Offshore Application," *SPE Production and Facilities* (2000) 15, 115-122.
15. Brie A., Pampuri F., Marsala A.F., and Meazza O. "Shear Sonic Interpretation in Gas-Bearing Sands," paper SPE 30595, presented at the SPE Annual Technical Conference & Exhibition, Dallas, USA (October 22-25, 1995).
16. Carlson, F. M. "Simulation of Relative Permeability Hysteresis to the Non-Wetting Phase," paper SPE 10157, presented at the SPE Annual Technical Conference & Exhibition, San Antonio, Texas, USA (October 5-7, 1981).
17. Carter, R. D. and Tracy, G. W. "An Improved Method for Calculating Water Influx," *Transactions of AIME* (1960) 219, 215-417.
18. Chang, Y-B, Coats B. K., and Nolen, J. S. "A Compositional Model for CO₂ Floods Including CO₂ Solubility in Water," paper SPE 35164, *SPE Reservoir Evaluation & Engineering* (1998) 1 No. 2, 155-160; also presented at the Permian Basin Oil and Gas Recovery Conference, Midland, Texas, USA (1996).
19. Charbeneau, Randall J. *Groundwater Hydraulics and Pollutant Transport*, Englewood Cliffs, New Jersey, USA, Prentice Hall (2000).
20. Cheng, L., Reme, A.B., Shan. D., Coombe, D.A., and Rossen, W.R. "Simulating Foam Processes and High and Low Foam Qualities," paper SPE 59287, presented at the SPE/DOE Symposium on Improved Oil Recovery, Tulsa, Oklahoma, USA (April 3-5, 2000).
21. Cheshire, I. M. and Pollard, R. "Recent Advances in Nested Factorization," Proceedings of the Cambridge Conference on Mathematical Techniques for the Oil Industry (1987).
22. Clarkson, C.R., Pan, Z., Palmer, I. and Harpalani, S. "Predicting Sorption-Induced Strain and Permeability Increase With Depletion for Coalbed-Methane Reservoirs", SPE 114778-PA, *SPE Journal* (2010) 15, No. 1, 152–159.
23. Clifford, P.J., and Sorbie, K.S., "The Effect Of Chemical Degradation on Polymer Flooding," paper SPE 13586, presented at the SPE Oilfield and Geothermal Chemistry Symposium, Phoenix, Arizona, USA (March 9-11 1985).
24. Coats, K. H. "An Equation of State Compositional Model," paper SPE 8284, *Society for Petroleum Engineers Journal* (1980) 20, 363-376; also presented at the SPE 54th Annual Technical Conference and Exhibition, Las Vegas, Nevada, USA (September 23-26, 1979).
25. Coats, K. H. "In-Situ Combustion Model," *Society of Petroleum Engineers Journal* (1980) 533-554.
26. Coats, K. H. "Simulation of Gas Condensate Reservoir Performance," paper SPE 10512, presented at the Sixth SPE Symposium on Reservoir Simulation, New Orleans, Louisiana, USA (February 1-3 1982).
27. Coats, K.H. "IMPES Stability: The Stable Step," paper SPE 69225, presented at the SPE Reservoir Simulation Symposium, Houston, Texas, USA (February 11-14 2001).
28. Cook, R. E., Jacoby, R. H., and Ramesh, A. B. "A Beta-type Reservoir Simulator for Approximating Compositional Effects During Gas Injection" paper SPE 4272, *Society of Petroleum Engineers Journal* (1974) 14, No. 5, 471-481.
29. Crookston, H. B., Culham, W. E., and Chen, W. H. "Numerical Simulation Model for Thermal Recovery Processes," *Society of Petroleum Engineers Journal*, Transactions of AIME, (1979) 267, 37-58.

30. Crowe, C. M. and Nishio, M. "Convergence Promotion in the Simulation of Chemical Processes - The General Dominant Eigenvalue Method," *American Institute of Chemical Engineering Journal* (1975) 23, No. 3, 528-533.
31. Dake, L.P. *Fundamentals of Reservoir Engineering*, Amsterdam, The Netherlands, Elsevier Science BV (1978) 256.
32. de Montleau, P., Cominelli, A., Neylon, K., Rowan, D., Pallister, I., Tesaker, O., and Nygard I. "Production Optimization under Constraints Using Adjoint Gradients," Presented at 10th European Conference on the Mathematics of Oil Recovery, Amsterdam, The Netherlands, (September 4-7, 2006).
33. Dennis, J. E. and Schnabel, R. B. *Numerical Methods for Unconstrained Optimisation and Nonlinear Equations*, Classics in Applied Mathematics (unabridged and corrected ed.) Society for Industrial and Applied Mathematics (1996), first published by Prentice-Hall, Englewood Cliffs, New Jersey, USA (1983).
34. Edmister, W. C. and Lee, B. I. *Applied Hydrocarbon Thermodynamics, Volume I*, (2nd ed.), Houston, Texas, USA, Gulf Publishing Company (1984).
35. Ejioogu, G. C. and Fiori, M. "High-Pressure Saturated-Steam Correlations," *Journal of Petroleum Technology* (1987) 39, No. 12, 1585-1590.
36. Ewing, R. E., Boyett, B. A., Babu, D. K., and Heinemann, R. F. "Efficient Use of Locally Refined Grids for Multiphase Reservoir Simulation," paper SPE 18413, *Proceedings of the Tenth SPE Symposium on Reservoir Simulation*, (1989) 55-70; also presented at the SPE Symposium on Reservoir Simulation, Houston, Texas, USA, (February 6-8, 1989).
37. Farouq Ali, S. M. *Oil Recovery by Steam Injection*, Bradford, PA, USA, Producers Publishing Co., (1970).
38. Fenghour, A., Wakeham, W.A., and Vesovic, V. "The Viscosity of Carbon Dioxide," *Journal of Physical and Chemical Reference Data* (1998) 27, No.1, 31-44.
39. Fetkovich, M. J. "A Simplified Approach to Water Influx Calculations - Finite Aquifer Systems," *Journal of Petroleum Technology*, (1971) 23, No. 7, 814-828.
40. Garmeh, R., Izadi, M., Salehi, M., Romero, J. L., Thomas, C. P. and Marque, E. J. "Thermally Active Polymer to Improve Sweep Efficiency of Water Floods: Simulation and Pilot Design Approaches", paper SPE 144234: presented at the SPE Enhanced Oil Recovery Conference, Kuala Lumpur, Malaysia (July 19-21, 2011).
41. Gill, P.E., Murray, W., and Wright, M.H. *Practical Optimization*, London and New York, Academic Press (1981).
42. Gilman, J. R. and Kazemi, H. "Improved Calculation for Viscous and Gravity Displacement in Matrix Blocks in Dual-Porosity Simulators," paper SPE 16010 (includes a number of associated papers), *Journal of Petroleum Technology* (1988) 40, No. 1, 60-70.
43. Grabowski, J. W. and Rubin, B. "A Preliminary Numerical Simulation Study of In-situ Combustion in a Cold Lake Oil Sands Reservoir," *Journal of Canadian Petroleum Technology* (1981) 20, No. 2, 79-89.
44. Gunasekera, D., Childs, P., Cox, J., and Herring, J. "A Multi-Point Flux Discretization Scheme for General Polyhedral Grids," paper SPE 48855, presented at the International Oil & Gas Conference and Exhibition, Beijing, China (November 2-6, 1998).
45. Haaland, S.E. "Simple and explicit formulas for the friction factor in turbulent pipe flow," *Transactions of the ASME, Journal of Fluids Engineering* (1983) 105, No. 1, 89-90.

46. Hagedorn, A. R. and Brown, K. E. "Experimental Study of Pressure Gradients Occurring During Continuous Two-Phase Flow in Small-Diameter Vertical Conduits," *Journal of Petroleum Technology* (1965) 17, No. 4, 475-484.
47. Han G., Dusseault, M. "Description of fluid flow around a wellbore with stress-dependent porosity and permeability," *Journal of Petroleum Science and Engineering* (2003) 40, No. 1, 1-16.
48. Harmathy, T. Z. "Velocity of Large Drops and Bubbles in Media of Infinite and Restricted Extent," *American Institute of Chemical Engineering Journal* (1960) 6, No. 2, 281-288.
49. Hasan, A. R. and Kabir, C. S. "A Simplified Model for Oil/Water Flow in Vertical and Deviated Wellbores," paper SPE 54131, *SPE Production & Facilities* (1999) 14 No. 1, 56-62; also presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA, (September 27-30, 1998).
50. Hasan, A. R. and Kabir, C. S. "A Study of Multiphase Flow Behavior in Vertical Wells," paper SPE 15138, *SPE Production Engineering* (1988) 3, No. 2, 263-272.
51. Haywood, R. W. *Thermodynamic Tables in SI (Metric) Units* (3rd ed.), Cambridge, UK, Cambridge University Press (1990).
52. Hegre, T. M., Dalen, V., and Henriquez, A. "Generalised Transmissibilities for Distorted Grids in Reservoir Simulation," paper SPE 15622, presented at the SPE Annual Technical Conference and Exhibition, New Orleans, Louisiana, USA (October 5-8, 1986).
53. Henderson, G.D., Danesh, A., Tehrani, D.H., and Al-Kharusi, B. "The Relative Significance of Positive Coupling and Inertial Effects on Gas Condensate Relative Permeabilities at High Velocity," paper SPE 62933, presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, USA (October 1-4 2000).
54. Holmes, J. A. "Enhancements to the Strongly Coupled, Fully Implicit Well Model: Wellbore Crossflow Modeling and Collective Well Control," paper SPE 12259, presented at the SPE Symposium on Reservoir Simulation, San Francisco (November 15-18, 1983).
55. Holmes, J.A., Barkve, T., and Lund, Ø. "Application of a Multisegment Well Model to Simulate Flow in Advanced Wells," paper SPE 50646, presented at the SPE European Petroleum Conference, The Hague, The Netherlands, (October 20-22, 1998).
56. Hustad, O.S. "A Coupled Model for Three-Phase Capillary Pressure and Relative Permeability," paper SPE 74705, *Society of Petroleum Engineers Journal* (2002) 7, No. 1, 59-69.
57. Hustad, O. S. and Browning, D. J. "A Fully Coupled Three-Phase Model for Capillary Pressure and Relative Permeability for Implicit Compositional Reservoir Simulation," paper SPE 125429, presented at the SPE/EAGE Reservoir Characterization and Simulation Conference, Abu Dhabi, UAE (October 19-21, 2009).
58. Hustad, O.S. and Hansen, A.G. "A Consistent Formulation for Three-Phase Relative Permeabilities and Phase Pressures based on Three Sets of Two-Phase Data," *RUTH: A Norwegian Research Program on Improved Oil Recovery, Program Summary, Norwegian Petroleum Directorate* (1996) 183-194; also in *Proceedings of the Eighth European Symposium on Improved Oil Recovery*, (1995) 1 289.
59. Hustad, O. S. and Holt, T. "Gravity Stable Displacement of Oil by Hydrocarbon Gas After Waterflooding," paper SPE 24116, presented at the SPE/DOE Eighth Symposium on Enhanced Oil Recovery, Tulsa, Oklahoma, USA (April 22-24 1992).
60. Hustad, O.S., Kløv, T., Lerdahl, T. R., Berge, L.,I., Stensen, J., Å., and Øren, P-E. "Gas Segregation During WAG Injection and the Importance of Parameter Scaling in Three-Phase Models," paper SPE

- 75138, presented at the SPE /DOE Improved Oil Recovery Symposium, Tulsa, Oklahoma, USA (April 13-17, 2002).
61. Infochem, 2019. Multiflash User Guide for Models and Physical Properties. Infochem/KBC Advanced Technologies Ltd., Version 7.1 (2019).
62. Kazemi, H., Merrill JR., L. S., Porterfield, K. L., and Zeman, P. R. "Numerical Simulation of Water-Oil Flow in Naturally Fractured Reservoirs," paper SPE 5719, *Society of Petroleum Engineers Journal* (1976) 16, No. 6, 317-326.
63. Kell, G.S and Whalley, E. "Reanalysis of the density of liquid water in the range 0-150 C and 0-1 kbar," *Journal of Chemical Physics* (1975), 62 No. 9, 3496-3503.
64. Kenyon, D. E. and Behie, A. "Third SPE Comparative Solution Project: Gas Cycling of Retrograde Condensate Reservoirs," paper SPE 12278, *Journal of Petroleum Technology* (1987) 39, No. 8, 981-997.
65. Killough, J. E. "Reservoir Simulation with History-dependent Saturation Functions," paper SPE 5106, *Society of Petroleum Engineers Journal* (1976) 16, No. 1, 37-48.
66. Kovscek, A.R. "Reservoir Simulation of Foam Displacement Processes," Presented at the 7th UNITAR International Conference on Heavy Crudes and Tar Sands, Beijing, China (October 27-31, 1998).
67. Land, C. E. "Calculation of Imbibition Relative Permeability for Two- and Three-Phase Flow from Rock Properties," paper SPE 1942, *Society of Petroleum Engineers Journal* (1968) 8, No. 2, 149-156.
68. Landau, L.D. and Lifshitz, E. M. *Theory of Elasticity: Volume 7 (Course of Theoretical Physics)* (3rd ed.), Oxford, UK, Butterworth-Heinemann (1984).
69. Leibovici, C.F., Barker, J.W., and Waché, D. "A Method for Delumping the Results of a Compositional Reservoir Simulation," paper SPE 49068, presented at the SPE Annual Technical Conference & Exhibition, New Orleans, Louisiana, USA (September 27-30, 1998).
70. Lorentz, J., Bray, B. G., and Clark, C. R. J. "Calculating Viscosity of Reservoir Fluids from their Composition," *Journal of Petroleum Technology* (1964) 1171, 231
71. Martin, J. J. "Cubic Equations of State-Which?" *I and EC Fundamentals*, (1973) 18, 81.
72. Mavko, G., Mukerji, T., and Dvorkin, J. *The Rock Physics Handbook: Tools for Seismic Analysis in Porous Media*, Cambridge UK, Cambridge University Press (1998).
73. Mavor, M. J. and Gunter, W. D. "Secondary Porosity and Permeability of Coal vs. Gas Composition and Pressure," paper SPE 90255, *SPE Reservoir Evaluation & Engineering* (2006) 9, No. 2, 114-125.
74. Michelsen, M. L. "The isothermal flash problem. Part I. Stability," *Fluid Phase Equilibria* (1982) 9, 1-19.
75. Mott, R. "Calculating Well Deliverability in Gas Condensate Reservoirs," Presented at the 10th European Symposium on Improved Oil Recovery, Brighton, UK (August 18-20 1999).
76. Nghiem, L. X., Collins D. A., and Sharma, R. "Seventh SPE Comparative Solution Project: Modeling of Horizontal Wells in Reservoir Simulation," paper SPE 21221, presented at the SPE Symposium on Reservoir Simulation, Anaheim, California, USA (February 17-20, 1991).
77. Pal, R. and Rhodes, E. "Viscosity/Concentration Relationships for Emulsions," *Journal of Rheology* (1989) 33, No. 7, 1021-1045.
78. Palmer, I. and Mansoori, J. "How Permeability Depends on Stress and Pore Pressure in Coalbeds: A New Model," paper SPE 52607, *SPE Reservoir Evaluation & Engineering* (1998) 1, No. 6, 539-544.

79. Patillo, P.D., Kristiansen, T.G., Sund, G.V., and Kjelstadli, R.M. "Reservoir Compaction and Seafloor Subsidence at Valhall," paper SPE/ISRM 47274, presented at the Eurock '98 meeting, Trondheim, Norway (July 8-10, 1998).
80. Pedersen, K. S., Fredenslund, Aa., and Thomassen, P. *Properties of Oils and Natural Gases*, Contributions in Petroleum Geology and Engineering, Vol. 5, Houston, Texas, USA, Gulf Publishing Company (1989).
81. Pedrosa, O. A. and Aziz, K. "Use of Hybrid Grid in Reservoir Simulation," paper SPE 13507, *Society of Petroleum Engineers Journal* (1986) 1, No. 6, 611-621.
82. Perry, R. H. and Green, D.W. *Perry's Chemical Engineers' Handbook* (6th ed.), USA, McGraw Hill (1984).
83. Pierre Lebel, J. "Dynamic Fracture Modelling Approach for Cold Lake Cyclic Steam Stimulation," paper SPE 79010, presented at the SPE International Thermal Operations and Heavy Oil Symposium and International Horizontal Well Technology Conference, Calgary, Canada (November 4-7, 2002).
84. Ponting, D. K. "Corner Point Geometry in Reservoir Simulation" Proceedings of the Joint IMA/SPE European Conference on the Mathematics of Oil Recovery, Cambridge University, UK (July 25-27, 1989).
85. Ponting, D. K., Banks, D., Foster, B. A., Naccache, P. F., Nicholas, M. O., Pollard, R. K., Rae, J., and Walsh, S. K. "An Efficient Fully Implicit Simulator," *Society of Petroleum Engineers Journal* (1983) 23, 544-552.
86. Potter II, R. W., Babcock, R. S., and Brown, D. L. "A new method for determining the solubility of salt in aqueous solutions at elevated temperatures," *Journal of Research of the U.S. Geological Survey* (1977) 5, No. 3, 389-395.
87. Prats, M. *Thermal Recovery*, SPE Monograph Series (Vol. 7) Society of Petroleum Engineers (1982).
88. Pritchett, J., Frampton, H., Brinkman, J., Cheung, S., Morgan, J., Chang, K. T., Williams, D. and Goodgame J.: "Field Application of a New In-Depth Waterflood Conformance Improvement Tool", paper SPE 84897, presented at the *SPE International Improved Oil Recovery Conference in Asia Pacific, Kuala Lumpur, Malaysia* (October 20-21 2003).
89. Quandalle, P. and Sabathier, J. C. "Typical Features of a Multipurpose Reservoir Simulator," paper SPE 16007, *SPE Reservoir Engineering* (1989) 4, No. 4, 475-480.
90. Reid, R.C., Prausnitz, J. M. and Polling, B. E. *The Properties of Gases and Liquids*, New York, USA, McGraw-Hill (1987).
91. Rozon, B. J. "A Generalized Finite Volume Discretization Method for Reservoir Simulation," paper SPE 18414, presented at the SPE Symposium on Reservoir Simulation, Houston, Texas, USA (February 6-8 1989).
92. Rubin, B. and Blunt, M. J. "Higher-Order Implicit Flux Limiting Schemes for Black Oil Simulation," paper SPE 21222, presented at the SPE Symposium on Reservoir Simulation, Anaheim, California, USA (February 17-20, 1991).
93. Russell, D.G., Goodrich, J.H., Perry, G.E and Bruskotter, J.F "Methods of Predicting Gas Well Performance", *Transactions of the ASME, Journal of Petroleum Technology* (1966) 99-108.
94. Saad, Y. *Iterative Methods for Sparse Linear Systems*, Boston, USA, PWS Publishing Company (1996).
95. Shaw, G., and Stone, T. "Finite Volume Methods for Coupled Stress/Fluid Flow in a Commercial Reservoir Simulator," paper SPE 93430, presented at the SPE Reservoir Simulation Symposium, The Woodlands, Texas, USA (January 31 - February 2, 2005).

96. Sheng J. *Modern Chemical Enhanced Oil Recovery. Theory and Practice*, Burlington, MA, USA, Gulf Professional Publishing, Elsevier Inc.(2011).
97. Shi, H., Holmes, J.A., Diaz, L. R., Durlofsky, L. J., and Aziz, K. "Drift-Flux Parameters for Three-Phase Steady-State Flow in Wellbores," paper SPE 89836, *Society of Petroleum Engineers Journal* (2005) 10, No. 2, 130-137; also presented at the SPE Annual Technical Conference and Exhibition, Houston, Texas, USA (September 26-29, 2004).
98. Shi, H., Holmes, J.A., Durlofsky, L. J., Aziz, K., Diaz, L. R., Alkaya, B., and Oddie, G. "Drift-Flux Modeling of Two-Phase Flow in Wellbores," paper SPE 84228, *Society of Petroleum Engineers Journal* (2005) 10, No. 1, 24-33; also presented as "Drift-Flux Modeling of Multiphase Flow in Wellbores," at the SPE Annual Technical Conference and Exhibition, Denver, Colorado, USA (October 5-8, 2003).
99. Shiralkar, G. A., and Stephenson, R. E. "A General Formalism for Simulating Physical Dispersion and a New Nine-Point Scheme," paper SPE 16975, *SPE Reservoir Engineering* (1991) 6, No. 1, 115-120.
100. Singh, K. and Whitson, C.H. "Gas Condensate Pseudopressure in Layered Reservoirs," paper SPE 117930, presented at the Abu Dhabi International Petroleum Exhibition and Conference, Abu Dhabi, UAE, (November 3-6, 2008).
101. Sokolnikoff, I.S. and Redheffer, R.M. *Mathematics of Physics and Modern Engineering*, New York, USA, McGraw-Hill (1967).
102. Soreide, I., and Whitson, C.H. "Peng Robinson predictions for hydrocarbons, CO₂, N₂ and H₂S with pure water," *Fluid Phase Equilibria* (1992) 77, 217-240.
103. Spycher, N. and Pruess, K. "CO₂-H₂O mixtures in the geological sequestration of CO₂.II. Partitioning in chloride brines at 12-100 C and up to 600 bar," *Geochimica et Cosmochimica Acta* (2005) 69, No. 13, 3309-3320.
104. Spycher, N. and Pruess, K. "A Phase-Partitioning Model for CO₂-Brine Mixtures at Elevated Temperatures and Pressures: Application to CO₂-Enhanced Geothermal Systems," *Transport Porous Media* (2009) 82, No. 1, 173-196.
105. StatoilHydro ASA "Method and Apparatus for Determining Capillary Pressures in a Three Phase Fluid Reservoir," United States Patent Application No. 12/266307 (May 6, 2010).
106. Stone, H. L. "Estimation of Three-Phase Relative Permeability and Residual Oil Data," *Journal of Canadian Petroleum Technology* (1973) 12, No. 4, 53-61.
107. Stone, H. L. "Probability Model for Estimating Three-Phase Relative Permeability," paper SPE 2116, *Journal of Canadian Petroleum Technology* (1973) 22, No. 2, 214-218.
108. Stone, T.W., Bennett, J., Law, D.H.-S., and Holmes, J.A. "Thermal Simulation with Multisegment Wells," paper SPE 78131, *SPE Reservoir Evaluation & Engineering* (2002) 5, No. 3, 206-218.
109. Todd, M. and Longstaff, W. "The Development, Testing and Application of a Numerical Simulator for Predicting Miscible Flood Performance," paper SPE 3484, *Journal of Canadian Petroleum Technology* (1972) 24, No. 7, 874-882.
110. Tortike, W. S. and Farouq Ali, S. M. "Saturated Steam Property Functional Correlations for Fully Implicit Thermal Reservoir Simulation," paper SPE 17094, *Society of Petroleum Engineers Journal* (1989) 4, No. 4, 471-474.
111. Trangenstein, J. A. "Analysis of a Model and Sequential Numerical Method for Thermal Reservoir Simulation," *The Mathematics of Oil Recovery*, King, P.R. (ed.), Oxford, UK, Clarendon Press (1992), 359.

112. Valle, A. "Three Phase Gas-Oil-Water Pipe Flow," Ph.D. Thesis, Imperial College, London, UK (2000)
113. Van Everdingen A.F. and Hurst W. "The Application of the Laplace Transformation to Flow Problems in Reservoirs," paper SPE 949305, *Petroleum Transactions, AIME* (1949) 186, 305-324.
114. van Lingen, P., Sengul, M., Daniel, J. and Cosentino, L. "Single Medium Simulation of Reservoirs with Conductive Faults and Fractures," paper SPE 68165, presented at the SPE Middle East Oil Show, Bahrain (March 17-20, 2001).
115. Vesovic, V., Wakeham, W. A., Olchoway, G.A., Sengers, J.V., Watson, J.T.R. and Millat, J. "The Transport Properties of Carbon Dioxide," *Journal of Physical and Chemical Reference Data* (1990) 19, No.3, 763.
116. Vinsome, P. K. W. "Orthomin, an Iterative Method for Solving Sparse Banded Sets of Simultaneous Linear Equations," paper SPE 5729, presented at the SPE Symposium on Reservoir Simulation, Los Angeles, USA (February 19-20, 1976).
117. Vinsome, P.K.W. and Westerveld, J. "A simple method for predicting cap and base rock heat losses in thermal reservoir simulators," *Journal of Canadian Petroleum Technology* (1980) 19, No. 3, 87-90.
118. Wallis, G. B. *One-dimensional Two-phase Flow*, New York, USA, McGraw-Hill (1969).
119. Wallis, G. B. and Makkenchery, S. "The Hanging Film Phenomenon in Vertical Annular Two-phase Flow," *Journal of Fluids Engineering* (1974) 96, No. 3, 297-298.
120. Wallis, J. R. "Incomplete Gaussian Elimination as a Preconditioning for Generalised Conjugate Gradient Acceleration," paper SPE 12265, presented at the SPE Symposium on Reservoir Simulation, San Francisco, USA (November 15-18, 1983).
121. Wallis, J. R., Little, T. E., and Nolen, J. S.: "Constrained Residual Acceleration of Conjugate Residual Methods," paper SPE 13536 presented at the SPE Reservoir Simulation Symposium, Dallas, Texas, USA (February 10-13, 1985).
122. Weinstein, H.G., Chappellear, J.E., and Nolen, J.S. "Second Comparative Solution Project: A Three-Phase Coning Study," paper SPE 10489, *Journal of Petroleum Technology* (1986) 38, No. 3, 345-353.
123. Welty, J.R., Wicks, C.E., and Wilson, R.E. *Fundamentals of Momentum Heat and Mass Transfer*, New Jersey, USA, John Wiley and Sons (1984).
124. Whitson, C. H. and Fevang, Ø. "Generalised Pseudopressure Well Treatment in Reservoir Simulation," Presented at the IBC Technical Services Conference on Optimisation of Gas Condensate Fields, Aberdeen, UK (June 26-27, 1997).
125. Whitson, C. H. and Fevang, Ø. "Modeling Gas Condensate Well Deliverability," paper SPE 30714, *SPE Reservoir Engineering* (1996) 11, No. 4, 221-230; also presented at the SPE Annual Technical Conference and Exhibition, Dallas, Texas, USA (October 22-25, 1995).
126. Whitson, C.H., Fevang, O., and Saevareid, A. "Gas Condensate Relative Permeability for Well Calculations," paper SPE 56476, presented at the SPE Annual Technical Conference, Houston, Texas, USA (October 3-6 1999).
127. Wilson, G. "A Modified Redlich-Kwong EOS, Application to General Physical Data Calculations," paper 15C, presented at the American Institute of Chemical Engineers National Meeting, Cleveland, Ohio, USA (May 1968).
128. Wingard, J. S. "Multicomponent Multiphase Flow in Porous Media with Temperature Variation," PhD thesis, Stanford University, Palo Alto, California, USA (1988).

129. Yanosik, J. L. and McCracken, T. A. "A Nine-Point, Finite-Difference Reservoir Simulator for Realistic Prediction of Adverse Mobility Ratio Displacements," paper SPE 5734, *Society of Petroleum Engineers Journal* (1979) 19, No. 4, 253-262.
130. Zakirov, I.S., Aanonsen, S.I., Zakirov, E.S., and Palatnik, B.M. "Optimizing Reservoir Performance by Automatic Allocation of Well Rates," Presented at the European Conference on the Mathematics of Oil Recovery, Leoben, Austria (September 3-6 1996).
131. Zaytsev, I.D. and Aseyev, G.G. *Properties of Aqueous Solutions of Electrolytes*, Boca Raton, Florida, USA CRC Press (1993).
132. Zimmerman, R.W. *Compressibility of Sandstones*, New York, USA Elsevier Science (1991).
133. Zuber, N. and Findlay, J. A. "Average Volumetric Concentration in Two-Phase Flow Systems," *Transactions of ASME, Ser. C, Journal of Heat Transfer* (1965) 87, No. 4, 453-468.